INORGANIC CHEMISTRY

UNIT 5 NOTES

RADIOPHARMACEUTICALS

- MEASUREMENT OF RADIOACTIVITY
- PROPERTIES OF RADIATIONS
- HALF LIFE
- RADIOISOTOPES
- STORAGE CONDITIONS
- PRECAUTIONS

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RADIOACTIVITY

- Radioactivity is the spontaneous emission of radiation from the nucleus of an unstable atom.
- This generally occurs when an atom's nucleus has an imbalance of protons and neutrons, causing it to release energy in the form of particles or electromagnetic waves to reach more stable form.

RADIOACTIVE SUBSTANCES

- Radiactive substances are materials that contain unstable atomic nuclei which spontaneously decay, releasing radiations in the form of α, β & γ radiations.
- Example: Uranium - 238, Carbon - 14 etc.

RADIOPHARMACEUTICALS

- Radiopharmaceuticals are drugs that contain radiactive materials and are used for diagnosis and treating diseases.
- They are designed to target specific organs, tissues or cells in the body.
- Example: Iodine - 131 for thyroid cancer treatment.

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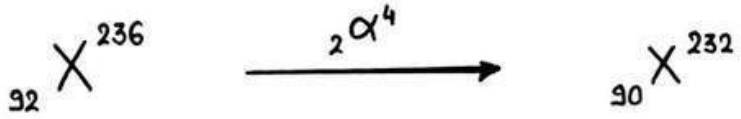
TYPES OF RADIATIONS

There are mainly 3 types of radiations emitted by radiactive substances:

① α - Rays
② β - Rays
③ γ - Rays

① α - RAYS

- These are positivily charged rays & consist of two protons and 2 neutrons.
- It have +2 charge due to absence of electron
- These are heavy, slow moving and their penetration power is slow.
- They have high ionization ability, hence ionize the gas through which they pass.
- During the emission of α- particle from a radioactive substance, atomic number decreased by 2 unit & mass number decreased by 4 units.

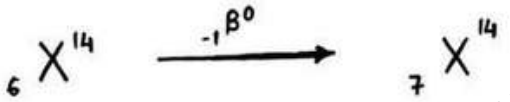


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② β - Rays

- These rays or particles are negatively charged.
- They have neglegible or varying mass.
- These are having smaller mass, higher speed and thus β particles are much more penetrating than α particles.
- They have lower ionizing power than α- rays.
- During the emission of β particles from a radioactive substance, atomic number increased by 1 unit & there is no change in mass number.



③ γ - Rays

- These rays are neutral i.e., do not carrying charge.
- They are pure energy with no mass & no charge.
- They travel at speed of light
- They have highest penetrating power.
- They have least ionizing power.

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RADIOISOTOPES

- Isotopes are the atoms of the same element that have the same number of protons but different number of neutrons i.e.. they have same atomic number but different atomic masses.
- Example : ${}_6C^{12}$, ${}_6C^{13}$, ${}_6C^{14}$.
- Radioisotopes are atoms that have an unstable nucleus & emit radiation as they decay into more stable form .
- They are widely used in medicine , industry, scientific research and nuclear power .
- Example , ${}^{3}H$, ${}^{14}C$, ${}^{131}I$ etc.

HALF LIFE

- Half Life is defined as the time in which the amount of radionuclide decays to half its initial value.
- During each half - life period , the amount of radioactive material decreased by 50 %
- The half life is a constant property for a given isotope and is not affected by temperature , pressure & chemical state.
- Example : The half life of carbon - 14 is 5,730 years , if you start with 100 grams of carbon - 14 , after 5,730 years only 50 gram will remain & so on.

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MEASURMENT OF RADIOACTIVITY

- Radioactivity is measured using different units and instruments, depending on the type of measurment being performed.
- The key instrument used for measurment of radioactivity is Grieger - Muller Counter.

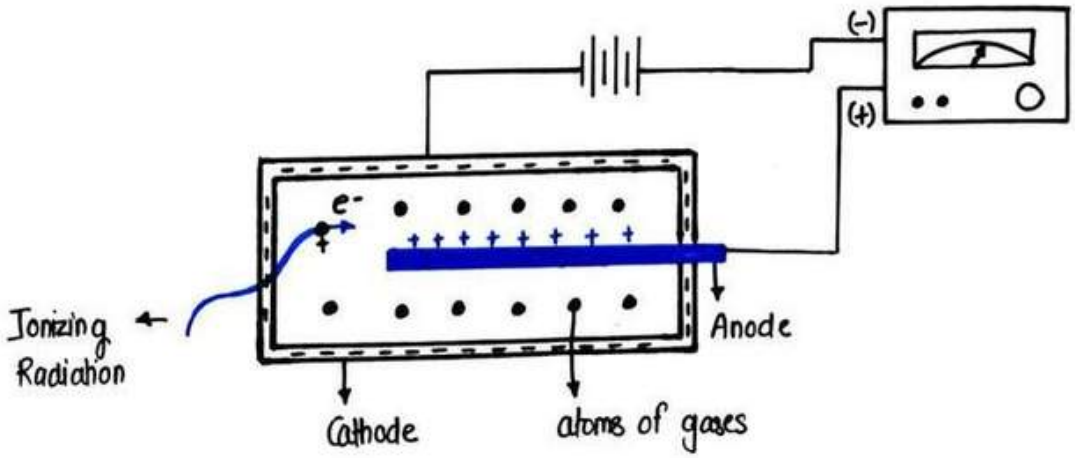
GIEGER - MULLER COUNTER

- A Grieger - Muller Counter is an instrument used to detect and measure ionizing radiations, such as alpha (α), beta (β) and gamma (γ) radiations.
- It operates on the ionization of gas within a specially designed tube when exposed to radiation.

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Construction

- It consist of a cylindrical metal or glass tube filled with a low pressure inert gas (eg. argon, neon or helium)
- Inside the tube, there is a central wire electrode (anode) and an outer conductive casing (cathode)



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WORKING

- When ionizing radiations enters the GM tube, it collides with gas molecules, knocking out electrons & creating ion pairs.
- The freed electrons are accelerated towards the positively charged anode, gaining energy and causing additional ionization in a chain reaction.
- This results in a brief electrical pulse due to the sudden flow of charge which is then processed & recorded by the circuit.

Advantages

- Detects low level of radiations.
- Portable & Easy to use
- Provides real time data.

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HANDLING & STORAGE OF RADIOPHARMACEUTICALS

- Radiopharmaceuticals are radioactive substances used in medical settings & treatments.
- Since they emit radiation, special care is needed to protect people, equipment and environment.

① HANDLING PRECAUTIONS

- Wear Protective Gear: Always use gloves, safety goggles, lab coats
- Limit Exposure Time: Work quickly & carefully to reduce exposure time.
- Maintain Distance: Use tools like forceps or tongs to handle materials instead of bare hands.
- Use Sheilding: Store & Handle materials in lead containers or behind protective screens.

- Label all containers with radiactive warning sign.
- Use syringe sheilds when injecting radiopharmaceuticals.
- Work in a well ventilated area to avoid inhaling radioactive gases.

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② STORAGE PRECAUTIONS

- Store in Lead Containers : Prevents radiation from escaping.
- keep Different Isotopes Separate : Avoids contamination between different materials.
- Control Temperature : Some radiopharmaceuticals need to be kept in refrigerators.
- Lock the storage Area : Only authorised personnel should have access.

- Use Geiger counter to check for radiation leaks.
- Check storage containers regularly for signs of leaks or damage.
- keep records of materials received, used & disposed of.

③ TRANSPORT PRECAUTIONS

- Use Proper Packaging : Materials should be in sheilded, labeled containers.
- Follow Safety Rules : Transportation must follow government regulations to prevent exposure to the public.
- Minimize Transport Time : Reduce unnecessary stops to limit radiation exposure.

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THANK YOU

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